

# Pancreatic Cancer Survival Analysis

MSK 2024 cohort · 270-day classification and survival modeling

UE24CS352A · Machine Learning Mini-Project

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## Problem and objective

We study whether clinical data can help estimate pancreatic cancer survival risk. The analysis follows the broad workflow of Jamalian's report and extends its Naive Bayes classifier with four additional classifiers. The Stanford cohort described in that report is restricted, so this project uses the public cBioPortal PDAC MSK 2024 cohort. Results describe MSK patients and do not reproduce the paper's cohort, radiomic features, or numerical scores.

## Dataset

The downloaded cBioPortal study contains 2,336 patients. We retain 2260 with known overall-survival time/status and positive follow-up (1521 observed deaths; 739 right-censored). Follow-up months are converted to days. Classification uses baseline age, sex, stage, genomic group, tumor location, tobacco exposure, and recorded medical histories. The Cox model uses age, sex, and stage. No radiomic features are included; treatment fields are omitted to avoid post-diagnosis predictors.

The classification target is a recorded death before 270 days. Patients censored before that cutoff are excluded because their status is unknown. This leaves 2027 known labels, including 533 high-risk events; 233 early-censored patients are omitted from this classifier task.

## Approach

We compare discretized Naive Bayes, Logistic Regression, RBF SVM, Random Forest, and Histogram Gradient Boosting. Numeric missing values are median-imputed and scaled; categorical values are most-frequent-imputed and one-hot encoded. Preprocessing is fitted inside each training fold. Fixed classifiers use stratified five-fold cross-validation (seed 229) and report balanced accuracy, precision, recall, F1, and ROC-AUC. Nested tuning uses a three-fold parameter search within each of five outer folds; held-out outer folds are used only for scoring. The penalized Cox model is evaluated with five-fold cross-validation and Harrell's C-index. SMOTE is omitted because synthetic interpolation can create invalid censored survival records.

# Results and Interpretation

MSK 2024 · five-fold evaluation · 270-day primary cutoff

## Fixed classifier comparison

| Classifier          | Bal. acc. | Precision | Recall | F1    | ROC-AUC |
|---------------------|-----------|-----------|--------|-------|---------|
| Logistic Regression | 0.653     | 0.389     | 0.696  | 0.499 | 0.692   |
| SVM (RBF)           | 0.652     | 0.390     | 0.685  | 0.497 | 0.666   |
| Naive Bayes         | 0.650     | 0.388     | 0.685  | 0.495 | 0.684   |
| Random Forest       | 0.590     | 0.346     | 0.552  | 0.425 | 0.636   |
| Gradient Boosting   | 0.540     | 0.374     | 0.201  | 0.261 | 0.631   |

Fixed Logistic Regression led balanced accuracy (0.653) and ROC-AUC (0.692). Its recall was 0.696 and precision 0.389, showing that many high-risk flags were false positives.

## Nested tuning and survival results

| Classifier          | Fixed bAcc | Tuned bAcc | Tuned ROC-AUC |
|---------------------|------------|------------|---------------|
| SVM (RBF)           | 0.652      | 0.661      | 0.675         |
| Random Forest       | 0.590      | 0.658      | 0.689         |
| Naive Bayes         | 0.650      | 0.652      | 0.682         |
| Logistic Regression | 0.653      | 0.650      | 0.692         |
| Gradient Boosting   | 0.540      | 0.537      | 0.664         |

Tuned RBF SVM had the highest balanced accuracy (0.661; ROC-AUC 0.675); tuned Random Forest reached 0.658. Tuning helped some classifiers and reduced scores for others. The clinical Cox model's mean C-index was 0.657 (SD 0.016; 5/5 successful folds).

## Naive Bayes horizon sensitivity

| Horizon  | Known labels | High risk | Bal. acc. |
|----------|--------------|-----------|-----------|
| 240 days | 2044         | 482       | 0.664     |
| 270 days | 2027         | 533       | 0.650     |
| 300 days | 2014         | 602       | 0.658     |
| 330 days | 1997         | 658       | 0.653     |
| 360 days | 1989         | 712       | 0.653     |

The exploratory MSK threshold sweep was highest for Naive Bayes at 240 days (0.664); the primary comparison remains at 270 days to follow the reference cutoff. This horizon sweep is not nested selection. The full-cohort median-risk Kaplan–Meier split had a descriptive log-rank p-value of 1.73e-72; it is not out-of-fold validation.

## Conclusion and limitations

These results describe one public MSK cohort and are not clinical decision support. The Stanford cohort and 900 radiomic features in the reference report are unavailable here, so the scores are not directly comparable. Early censoring reduces the classification sample, and the Kaplan–Meier split uses the full cohort. Report the cohort, 270-day label definition, exclusions, features, and nested evaluation whenever quoting results.

Sources: Jamalian, A. (2020), CS229 Project Final Report: Pancreatic cancer prognosis using clinical and radiomic data. cBioPortal PDAC MSK 2024 study: [https://www.cbioportal.org/study/summary?id=pdac\\_msk\\_2024](https://www.cbioportal.org/study/summary?id=pdac_msk_2024). Code, metrics, and run metadata are in the accompanying repository.